



M I C H A E L P E Ñ A

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SUMMARY

Master's trained statistician with a strong research background in biostatistics focused on physical therapy data science. Experienced with demonstrating core skills in computational statistics involving programming, exploratory data analysis, data mining and processing, and data visualization. Eager to collaborate with a driven research team aiming to protect the health of all Californians. Seeking Research Scientist roles in Sacramento County due to upcoming planned relocation.

EDUCATIONAL DEGREES & CERTIFICATES

California State University, Fullerton	May 2025
Masters of Science, Statistics	
California Polytechnic State University, Pomona	May 2023
Bachelors of Science, Applied Mathematics and Statistics	
IBM via edX.org	December 2023
Certificate of Completion, SQL for Data Science	
IBM via edX.org	June 2023
Certificate of Completion, Linux Commands & Shell Scripting Basics	

SKILLS

Programming	Software	Analytical Methods	Professionalism
Bash	Microsoft Excel	Statistical Interference	Data Visualization & Presentation
R	Microsoft Suite	Time Series Analysis	Independent Work
Python	Microsoft Copilot	Survival Analysis	Team Communication
SQL	STATA	Cross Validation	Project Management
VBA	R Shiny	Bayesian Data Analysis	Technical Writing

WORK HISTORY

Statistical Consultant, Dr. Marcel Lanza Lab University of Maryland School of Medicine, Remote	February 2025 – Present
• Collaborated with multidisciplinary research team of 3 fellow graduate researchers and 5 clinical researchers to ensure accurate data analysis for pilot study assessing fall risk in older adults • Conducted exploratory data analysis, applying appropriate statistical diagnostics such as Little's Missing Completely at Random test to assess missingness and inform imputation strategy • Developed and validated statistical model, such as k-nearest neighbor classification model, in powerful statistical software for quantitative data interpretation • Prepared academic writing for the statistical analysis sections and discussion sections of a research manuscript currently in preparation • Presented findings during stakeholder briefings, delivering data-driven recommendations to guide project direction	
Tutor Expert Mount San Antonio College, Walnut, California	November 2023 – Present
• Guided 100+ students through the process of data collection, cleaning datasets to remove inconsistencies, and data interpretation as part of their coursework and projects • Assisted 100+ students in analyzing quantitative data, interpret results, and provided feedback for written reports • Provided assistance in guiding students through statistical analysis methodologies and data visualization techniques using appropriate analytical software • Instructed 100+ students in drafting, reviewing, and editing of technical reports, ensuring that their findings were accurately conveyed	



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WORK HISTORY (CONTINUED)

Intern, New Student Programs & Family Engagement | Cal Poly Pomona, California

July 2023–October 2023

- Created clear and detailed itineraries and contact sheets in Microsoft Excel for multiple university orientations, ensuring smooth event coordination with attention to accuracy.
- Delivered impactful presentations to large groups of family members and prospective students, educating them about the university and its resources, demonstrating strong presentation skills.

Student Assistant & Test Proctor, Disability Resource Center | Cal Poly Pomona, California

August 2021–May 2023

- Applied effective interpersonal communication rules and guidelines with hundreds of students who had learning disabilities.
- Created in-depth protocols by writing guidelines for future student assistants to follow

PERSONAL AND ACADEMIC PROJECTS

Blood Types Distribution | Personal Project

October 2025

- Modeled the distribution of human blood types over 17 Mexican states using a complex hierarchical Bayesian sampling technology, using academic research literature

Electromyogram Prediction | Academic Project, Time Series Analysis

July 2024

- Built and presented a model to predict the type of active muscle, trained from 2.4+ million observed biostatistical muscle electromyogram datapoints

Transit Oriented Community Housing Costs | Academic Project, Statistical Learning

Fall 2024

- Demonstrated relationship between delayed construction and affordable housing by applying Statistical Survival Analysis to 300+ datapoints, improving methods from previous academic literature

Building 8, Elevator Wait-Time Prediction | Academic Project, Statistical Consulting

Spring 2023

- Designed a statistical model that accurately predicts the wait time of an elevator using on-site quantitative data collection

RELATED COURSEWORK

Statistical Inference and Probability

- Fundamentals of probability and mathematical intuition for statistics and statistical analysis

Statistical Computation

- Application of algorithms and simulations using software programming

Categorical Data Analysis

- Training of statistical models to accurately predict qualitative outcomes.

Time Series Analysis

- Application of training time dependent predictive models.

Statistical Learning

- Reviewing the machine learning and predictive training aspects of statistics

Bayesian Data Analysis

- Bayesian statistical methodologies with emphasis in biological and life science applications

RESEARCH SUMMARY

As part of my Master's Statistical Consulting Capstone, I worked under clinical researcher Dr. Marcel Lanza for a pilot study examining fall risk in older adults. The study is focused on evaluating quantitative data related to age-related differences in stepping kinematics and balance recovery to investigate how muscle structure influences movement outcomes. Using kinematic assessments and ultrasound imaging, I conducted exploratory data analysis, performed data imputation, and built data visualizations to support trend identification and risk factor analysis across age groups. To manage high-dimensional data, I applied Principal Component Analysis (PCA) for data reduction, transforming complex inputs into meaningful analytical components. I then computed a classification model using K-Nearest Neighbors (KNN), and applied cross-validation to achieve model optimization and strong performance metrics. The resulting, publication-ready model achieves the 5-Nearest Neighbor representation and clarifies the relationships between age, muscle structure, and stepping performance.